



16/20

Habib University

CS-330 Computer Architecture – Fall 2025

Instructor: Dr. Shehzad Hasan

Time = 40 minutes

Quiz 01

Max Points: 20

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.
- Answer sheets should contain all steps, working, explanations, and assumptions.
- Attempt the quiz on clean papers with black/blue ink.
- Print your name and HU ID on all sheets.
- Turn in your question paper along with your answer sheets.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	✓
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	

Question 1 [4 points]:

Compilers can have a profound impact on the performance of an application. Suppose there are two compilers A and B

- a. Assume that for an application, compiler-A generates a code having 5×10^9 dynamic instructions which are executed in 9.78 s, while compiler-B generates a code having 2.35×10^9 dynamic instructions with an execution time of 4.15 s on the same processor. Find the average CPI for each program given that the processor has a clock rate of 1.1 GHz. (2 marks)

$$A; \frac{5 \times 10^9}{9.78} = 51247443.8 \text{ instructions/second}$$

$$CPI = \frac{1.1 \times 10^9}{51247443.8} = 2.1516$$

$$CPI_A = 2.1516$$

$$B; \frac{2.35 \times 10^9}{4.15} = 566265060.2 \text{ instructions/second}$$

$$CPI = \frac{1.1 \times 10^9}{566265060.2} = 1.9426$$

$$CPI_B = 1.9426$$

- b. Assume the compiled programs now run on two different processors (compiler-A's code on P1 and compiler-B's code on P2). If the execution times on the two processors are the same, which processor's clock is faster and by how much? (2 marks)

$$\frac{2.1516 \times 5 \times 10^9}{\text{clock}_A} = \frac{1.9426 \times 2.35 \times 10^9}{\text{clock}_B}$$

$$\frac{\text{clock}_B}{\text{clock}_A} = \frac{1.9426 \times 2.35 \times 10^9}{2.1516 \times 5 \times 10^9} = \frac{1}{0.4243} = 2.3566$$

clock_A is faster by 2.36 times faster.

Question 2 [4 points]:

Consider the following program

```

addi x4, x0, 1000
addi x5, x0, 100
loop: bge x5, x4, exit
      slli x6, x5, 2
      lwu x7, 0(x6)
      sw x7, 100(x6)
      addi x5, x5, 4
      jal x0, loop
exit:
    
```

exit:

Let the clock cycle counts for each instruction type be as follows:

Arithmetic: 2 cycles
 Logical: 1 cycle
 Memory: 4 cycles
 Conditional branching: 3 cycles
 Unconditional branching: 2 cycles

Calculate the total number of clock cycles required to execute the program based on the instruction distribution. Let the instructions in the loop run 25 times, the loop terminating instruction, however, is going to run one more time.

$$5 \times 25 = 125 + 26 = 151 + 2 = 153 = I.C$$

$$\text{add} = 3 = 2 \times 25 = 50 + 2 = 27$$

$$\text{logical} = 1 = 25$$

$$\text{Memory} = 2 \times 25 = 50$$

$$\text{Unconditional} = 25$$

$$\text{Conditional} = 26$$

3.2

$$2 \times \frac{27}{153} + 1 \times \frac{25}{153} + 4 \times \frac{50}{153} + 3 \times \frac{25}{153} + 2 \times \frac{26}{153} = \frac{406}{153} \div 153$$

$$\text{CPU clock cycle} = 2.6536$$

Question 3 [4 points]:

Consider a computer running a program that requires 350s, with 140s spent executing memory instructions, and 120s spent executing arithmetic instructions.

- a. What would be the speedup achieved if the memory unit is enhanced by a factor of 5. (2 marks)

$$140 - \left(\frac{140}{5} \right) = 132s + 120 = 252s + 90 = 342s$$

$$140 - 28 = 112s + 120 = 232s + 90 = 242s$$

$$\frac{343}{350} = 1.02 \text{ times will be the speed up.}$$

$$\frac{350}{242} = 1.44 \text{ times will be the speed up.}$$

- b. Can the total program time be reduced by 20% by reducing only the time for arithmetic instructions? If yes, then by what factor should the arithmetic unit be improved? (2 marks)

$$\frac{20}{100} \times 350 = 70. \text{ Yes it can be reduced just by reducing}$$

arithmetic.

$$\frac{120}{70} = 1.71$$

arithmetic instructions need to be reduced by a factor of 1.71.

Question 4 [4 points]:

- a. Assume that array A is an array of 100 double words, x20 contains the base address of array A, and x21 corresponds to variable D. Compile the following C expression into its equivalent RISC-V assembly instructions (2 marks)

$$A[100] = D + A[18] - 30$$

```

li x18, 30
li x20, 0x100
li x21, 0 # D variable
li x19, 0
sllw x22, 18(x20)
add x22, x21, x22
sub x22, x22, x18
lw x22, 100(x20)

```

b. Consider now array A is an unsigned array of 100 halfwords. Make the required changes in the RISC-V assembly instructions for the same C expression. (2 marks)

```

li x18, 30
li x19, 0
li x20, 0x100
li x21, 0
lh x22, 18(x20)
add x22, x22, x21
sub x22, x22, x18
sh x22, 100(x20)

```

Question 5 [4 points]:

Encode the following RISC-V instructions to machine language code

a. lhu x9, -1396(x27)

Imm [11:0]															
funct7				rs2				rs1				funct3			
1	0	1	0	1	0	0	0	1	1	0	0	1	0	0	1

b. sll x19, x21, x20

funct7				rs2				rs1				funct3			
0	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0